AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

T318
Applied Network Security

Final Examination
Spring Semester 2025/2026

Date: 11/06/2026

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Number of Exam Pages: 4 (including this cover sheet)	**Time Allowed:** 3 Hours

Instructions:

- Total Marks: **50**
- This exam consists of **3** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are not allowed.

PART 1: MULTIPLE CHOICE QUESTIONS [10 Marks]

Question 1: (10 marks)

1. Which security property ensures that information is not disclosed to unauthorized users?
A. Integrity
B. Availability
C. Confidentiality
D. Authenticity

2. A weakness that can be exploited by a threat is known as a:
A. Attack
B. Vulnerability
C. Risk
D. Exposure

3. The Caesar cipher is an example of which technique?
A. Transposition
B. Substitution
C. Stream cipher
D. Block cipher

4. Which AES step provides non-linearity using an S-box?
A. ShiftRows
B. MixColumns
C. AddRoundKey
D. SubBytes

5. Which mode of operation encrypts each block independently?
A. CBC
B. CFB
C.OFB
D. ECB

6. Which principle most effectively reduces the impact of a compromised subsystem in a layered architecture?
A. Least Privilege
B. Isolation
C. Open Design
D. Psychological Acceptability

7. Which attack on 3DES reduces its effective security from 168 bits to about 112 bits?
A. Brute force
B. Chosen-plaintext
C. Meet-in-the-middle attack
D. Replay attack

8. Which attack surface reduction technique focuses on minimizing shared resources?
A. Layering
B. Encapsulation
C. Least Common Mechanism
D. Separation of Privilege

9. Which cipher is provably secure under information-theoretic assumptions?
A. Vigenère
B. DES
C. AES
D. One-time pad

10. Which security design principle is violated when a system hides its internal logic to appear secure?
A. Least Privilege
B. Fail-Safe Defaults
C. Open Design
D. Modularity

PART 2: SHORT ANSWERS QUESTIONS [22 Marks]

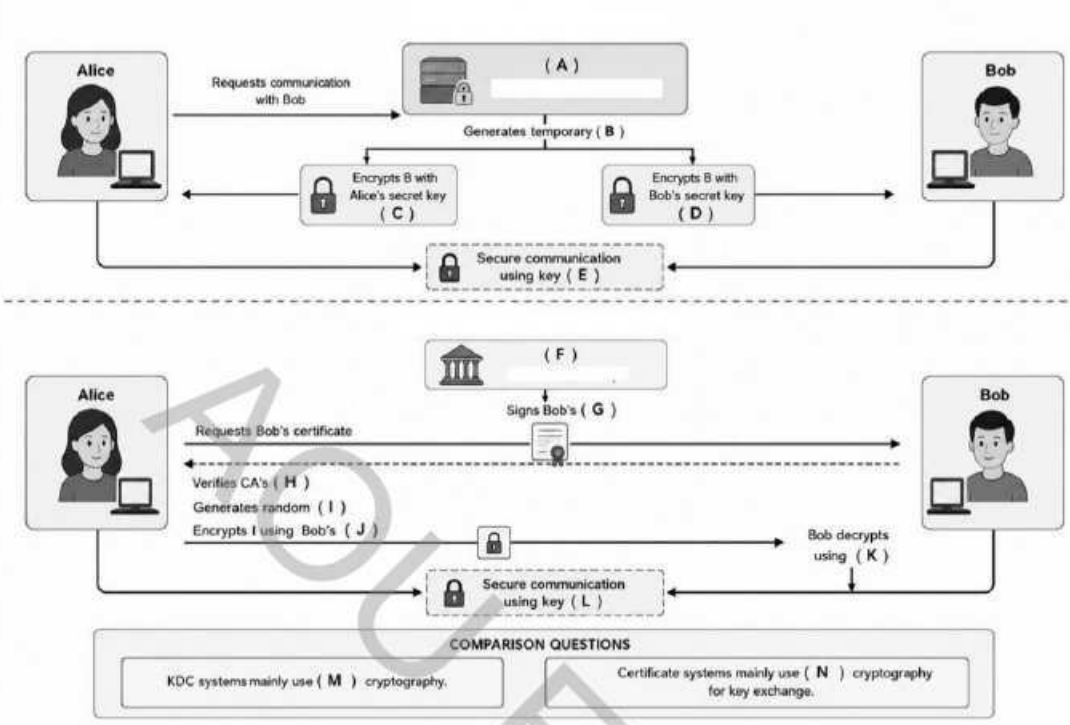
Question 2: (10 marks: 0.75 for each + 0.5)

Two organizations use different methods for secure key distribution:

- System 1 uses a Key Distribution Center (KDC)
- System 2 uses Digital Certificates and PKI

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Complete the missing labels (A → N) from the following word bank: Public Key Certificate, Certificate Authority, Session Key, Symmetric, Asymmetric, Public Key, Private Key, Shared Secret Key, CA Signature, KDC, Bob's Public Key, Bob's Private Key, Temporary Session Key}



Explain why a digital certificate must be signed by a trusted Certificate Authority.

Question 3: (12 marks: 6-6)

1. Complete the table below (0.5 for each).

Security Mechanism	Main Purpose	Uses Secret Key?	Provides Integrity?	Provides Authentication?
Hash Function	.....	.....	.....	.....
Collision-Resistant Hash	.....	.....	.....	.....
MAC	.....	.....	.....	.....

2. Complete the table below (1 for each).

Concept	Description
Salt	.....
Pepper	.....
Rainbow Table	.....
Brute-Force Attack	.....
Dictionary Attack	.....
Hash Stretching	.....

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PART 3: PROBLEM SOLVING [18 Marks]

Question 4: (8 marks: 1 for each)

A company allows employees to remotely access confidential company files through the internet. The network administrator deploys the following security mechanisms: TLS to secure web communications, VPN for remote employee access, IDS to monitor suspicious network activity and MLP (Multilevel Security Policy) to restrict access to classified documents. One employee accidentally attempts to access documents above their authorization level while connected remotely. At the same time, an attacker tries to intercept traffic and launch multiple logins attempts against the company server. (**provide brief and clear answers**).

1. Which technology encrypts the communication between the employee and the server?
2. Which technology creates a secure tunnel over the internet?
3. Which security mechanism detects suspicious login attempts?
4. Which mechanism prevents the employee from accessing higher-classified documents?
5. Which CIA security property is mainly provided by: 1) TLS, 2) IDS and 3) MLP
6. Explain how TLS and VPN differ.
7. Give one example of an attack that an IDS may detect.
8. Which access control model enforces "no write-down" to protect confidentiality?

Question 5: (10 marks: 2 for each)

Bob uses a toy RSA system with the following public key: n=33, e=7
Alice sends the ciphertext c=14
You are an attacker who wants to recover the message. Because the numbers are tiny, RSA is vulnerable.
1. Factor the modulus n
2. Compute Euler's totient $\varphi(n)$
3. Compute the private exponent d
4. Decrypt the ciphertext
5. Explain why factoring is the core difficulty of attacking RSA

End of questions

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